Assignment 1

Maths

Sol^n 1 Prove that U is subspace of R^n

$U = \{ x \in R^n : \sum_1^n x_i = 0 \}$.

Step 1: Setup subset property.

Given: $U = \{ x \in R^n : \sum_{i=1}^{n} x_i = 0 \}$ and $V = R^n$

1 proof: Since we have the clause $x \in R^n$ declared in the set definition, candidate vectors are already restricted in R^n. So, Nothing belongs of U without being in R^n.

Thus proved: $U \subseteq R^n$ (subset)

Step 2: Non Emptiness.

To prove: $0 \in U \Rightarrow U \neq 0$.

Assumptions: set of definition $U = \{ x \in R^n : \sum_{i=1}^{n} x_i = 0 \}$

proof: Since we have zero vector $0 = (0, 0, \dots 0)^T \in R^n$

$$\sum_{i=1}^{n} 0_i = 0 + 0 \dots + 0 = 0$$

It satisfies the defining summation of U.

Thus proved $0 \in U \Rightarrow U \neq 0$ (Non Empty set)

Step 3: Closure under vector Addition (inner op^n)

To prove : $x, y \in U \Rightarrow x+y \in U$

Assumption: let $x, y \in U$ be arbitrary vectors

proof : since we have $x \in U$ and $y \in U$

$$\Rightarrow \sum_{i=1}^{n} x_i = 0 \text{ and } \sum_{i=1}^{n} y_i = 0.$$

$$\Rightarrow 0+0=0$$

Since $\sum_{i=1}^{n} (x+y)_i = 0$, $x+y$ satisfies the defining condition of U.

Thus proved : $x+y \in U$ (closed under vector Addition)

Step 4: Closure under Scaler Multiplication (outer op^n)

To prove: ~~$\sum \lambda x_i$~~ $x \in U, \lambda \in R \Rightarrow \lambda x \in U$.

Assumption : let $x \in U$, arbitrary vectors & $\lambda \in R$ arbr scalar

proof : Since $\sum_{i=1}^{n} (\lambda x)_i = 0$:

$$\sum_{i=1}^{n} (\lambda x)_i = \sum_{i=1}^{n} \lambda x_i = \lambda \sum_{i=1}^{n} x_i = \lambda \cdot 0 = 0$$

Since $\sum_{i=1}^{n} (\lambda x_i) = 0$, the scaler vector λx satisfies the defining condition of U

Thus : $\lambda x \in U$

Hence overall conclusion.

Since we have

1. $U \subseteq R^n$

2. $U \neq \phi$ because $0 \in U$

3. $x, y \in U \Rightarrow x+y \in U$.

4. $x \in U, \lambda \in R \Rightarrow \lambda x \in U$.

$\therefore U$ is a subspace of R^n.

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 6 \\ 1 & 1 & 7 & 4 & 4 \\ 1 & 5 & 4 & 2 & 1 \end{pmatrix} \cdot b = \begin{pmatrix} 60 \\ 60 \\ 36 \end{pmatrix}$$

Sol^n 2

Step 1 Form Augumented Matrix $[A|b]$

$$[A|b] = \left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 1 & 1 & 7 & 4 & 4 & 60 \\ 1 & 5 & 4 & 2 & 1 & 36 \end{array}\right]$$

Step2: Reduce $[A|b]$ to REF.

$R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 0 & -1 & 4 & 0 & -2 & 0 \\ 0 & 3 & 1 & -2 & -5 & -24 \end{array}\right]$$

$R_3 \leftarrow R_3 + 3R_2$ and $R_2 \leftarrow (-1) R_2$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 1 & 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 13 & -2 & -11 & -24 \end{array}\right]$$ REF

Rank(A) = rank$([A|b])$ = 3 consistent

Rank(A) = 3 < 5 infinite many sol^n.

Basic = 3 $[x_1, x_2, x_3]$

Free = 2 x_4, x_5

Rank null Check.

nullity $(A) = n - \text{Rank}(A) = 5 - 3 = 2$

Step 3 Backward pass (RREF)

To Reduce REF

1. Apply $R_3 \leftarrow \frac{1}{13} R_3$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 6 & 60 \\ 0 & 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2/13 & -11/13 & -24/13 \end{array}\right]$$

2. Apply $R_1 \leftarrow R_1 - 3R_3$ and $R_2 \leftarrow R_2 + 4R_3$:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 58/13 & 111/13 & 852/13 \\ 0 & 1 & 0 & -8/13 & -18/13 & -96/13 \\ 0 & 0 & 1 & -2/13 & -11/13 & -24/13 \end{array}\right]$$

3. Apply $R_1 \leftarrow R_1 - 2R_2$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 74/13 & 147/13 & 1044/13 \\ 0 & 1 & 0 & -8/13 & -18/13 & -96/13 \\ 0 & 0 & 1 & -2/13 & -11/13 & -24/13 \end{array}\right]$$

Step 4 particular solm: x_p which satisfy $Ax_p = b$

Set free variable to $x_4 = 0$ $x_5 = 0$.

$$x_1 = \frac{1044}{13} \quad x_2 = -96/13, \quad x_3 = \frac{-24}{13}$$

$$x_p = \frac{1}{13}\begin{pmatrix} 1044 \\ -96 \\ -24 \\ 0 \\ 0 \end{pmatrix}$$

Step 5 : Homogenous system sol^n x_h.

$$Ax_h = 0$$

Take RREF & Express basic variable in term of free variables.

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & -74/13 & 147/13 & 1044/13 \\ 0 & 1 & 0 & -8/13 & -18/13 & -96/13 \\ 0 & 0 & 1 & -2/13 & -11/13 & -24/13 \end{array}\right]$$

$$x_1 = -\frac{74}{13}x_4 - \frac{147}{13}x_5$$

$$x_2 = \frac{8}{13}x_4 + \frac{18}{13}x_5$$

$$x_3 = \frac{2}{13}x_4 + \frac{11}{13}x_5$$

$$x_h = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -\frac{74}{13}x_4 - \frac{147}{13}x_5 \\ \frac{8}{13}x_4 + \frac{18}{13}x_5 \\ \frac{2}{13}x_4 + \frac{11}{13}x_5 \\ x_4 \\ x_5 \end{bmatrix}$$

$$\Rightarrow x_h = x_4 \begin{bmatrix} -\frac{74}{13} \\ \frac{8}{13} \\ \frac{2}{13} \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -\frac{147}{13} \\ \frac{18}{13} \\ \frac{11}{13} \\ 0 \\ 1 \end{bmatrix}$$

Absorb factor $\frac{1}{13}$ into arbitrary scalars $\lambda_1, \lambda_2 \in R$

$$h_1 = \begin{matrix} -74 \\ 8 \\ 2 \\ 13 \\ 0 \end{matrix}, \quad h_2 = \begin{matrix} -147 \\ 18 \\ 11 \\ 0 \\ 13 \end{matrix}$$

Thus $\quad x_h = \lambda_1 h_1 + \lambda_2 h_2 = \lambda_1 \begin{pmatrix} -74 \\ 8 \\ 2 \\ 13 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -147 \\ 18 \\ 11 \\ 0 \\ 13 \end{pmatrix}$

$\lambda_1, \lambda_2 \in R$

Step 6 General sol^n x_g & verification

$$x_g = x_p + x_h$$

Since $\cancel{A_1}$ $A x_p = b$ and $A x_h = 0$

$$x_g = x_p + x_h = A x_p + A x_h = b + 0 = b$$

$$x_g = A x_p + A x_h$$

$$= \frac{1}{13}\begin{pmatrix} 1044 \\ -96 \\ -24 \\ 0 \\ 0 \end{pmatrix} + h_1 \begin{pmatrix} -74 \\ 8 \\ 2 \\ 13 \\ 0 \end{pmatrix} + h_2 \begin{pmatrix} -147 \\ 18 \\ 11 \\ 0 \\ 13 \end{pmatrix}$$

$, h_1, h_2 \in R$

Verification

original Matrix A =

1	2	3	4	6	R_1
1	1	7	4	4	R_2
1	5	4	2	1	R_3

$h_1 = (1044; -96; -24)$

$h_2 = (-74, 8, 2, 13, 0)$

$h_3 = (-147, 18, 11, 0, 13)$

Now $R_1 \times h_1 = 1044 + 2(-96) + 3(-24) = 780/13 = 60$

$R_2 \times h_1 = 1044 + 1(-96) + 7(-24) = 780/13 = 60$

$R_3 \times h_1 = 1044 + 5(-96) + 4(-24) = 468/13 = 36$

Similarly $[R_1 \ldots R_3] \times h_2 = b$, $[R_1 \ldots R_3] \times h_3 = b$

$\underline{\text{Sol}^n}$ 3. Give $c = \begin{pmatrix} 1 & 2 & 1 & 2 & -1 \\ 1 & 1 & 0 & -1 & 4 \\ 1 & -2 & 4 & 1 & 0 \end{pmatrix}$

Step 0 REF Reduction.

$R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$

$\Rightarrow \begin{pmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & -4 & 3 & 1 & 1 \end{pmatrix}$

$R_3 \leftarrow R_3 - 4R_2$

$\Rightarrow \begin{pmatrix} 1 & 2 & 1 & 2 & -1 \\ 0 & -1 & -1 & -3 & 5 \\ 0 & 0 & 7 & 11 & -19 \end{pmatrix}$ REF

<u>part - I</u> column space col(c)

To prove $U = \text{col}(c)$ is a subspace of $\mathbb{R}^3$ then its basic & dim.

Step 1: To prove subset property $U \subseteq \mathbb{R}^3$

Given : $U = \text{span}\{c_1, c_2, c_3, c_4, c_5\}$
$= \{x \in \mathbb{R}^3 : \sum_1^5 a_i c_i\}$
$V = \mathbb{R}^3$

proof : Since we have $c_i \in \mathbb{R}^3$ and Vector space $\mathbb{R}^3$ is closed under $(+)$ & $(\cdot)$
Any linear combination $\sum_1^5 a_i c_i \in \mathbb{R}^3$ will fall in $\mathbb{R}^3$

Thus $U \subseteq \mathbb{R}^3$ subset property.

Step 2 Non-emptiness

To prove : $0 \in U \Rightarrow U \neq \phi$

Assumption : Definition of set $U = \text{Span}\{c_1 \cdots c_5\}$

proof : Since we have scalar coefficients .

$a_1 = a_2 = a_3 = a_4 = a_5 = 0$:

$$\sum_{i=1}^{5} a_i c_i = 0 \cdot c_1 + 0 \cdot c_2 \cdots + 0 c_5 = 0$$

The zero vector is expressible as a linear combination of $\{c_i\}$

Thus proved:

$0 \in U \Rightarrow U \neq \phi$ [Non zero vector]

Step3 Closure under Addition.

To prove : $x, y \in U \Rightarrow x+y \in U$

Assumption: $x, y \in U$, where

$x = \sum_{i=1}^{5} a_i c_i \; (a_i \in R)$ and $y = \sum_{i=1}^{5} b_i c_i \; (b_i \in R)$

Proof :

Since we have $x = \sum_{i=1}^{5} a_i c_i$ and $y = \sum_{i=1}^{5} b_i c_i$

$$\Rightarrow x+y = \sum_{i=1}^{5} a_i c_i + \sum_{i=1}^{5} b_i c_i$$

$$= \sum_{i=1}^{5} (a_i+b_i) c_i$$

$\hookrightarrow d_i \in R$ lets take for ease

$\Rightarrow$ So, Define $d_i = a_i + b_i$ $\because$ since $a_i, b_i \in R$ and R is closed under $+$, $d_i \in R$

$$x+y = \sum_{i=1}^{5} d_i c_i , \; d_i \in R$$

$\therefore$ The sum $x+y$ is expressible as linear combination of $\{c_i\}$ with real coefficients.

Thus proved $x+y \in U$ (Closed under vector Addition)

Step 4: Closure Under multiplication.

To prove : $x \in U, \lambda \in R \Rightarrow \lambda \cdot x \in U$

Assumption: Lets $x = \sum_{i=1}^{5} a_i c_i$ $(a_i \in R)$ as arbitrary vector & $\lambda \in R$ be an arbitrary scalar.

proof :

since we have $x = \sum_{i=1}^{5} a_i c_i$

$$\lambda x = \lambda \sum_{1}^{5} a_i c_i$$

$$= \sum_{i=1}^{5} (\lambda a_i) \cdot c_i$$

Define $e_i = \lambda a_i$. ~~Sh~~ since $\lambda, a_i \in R$ and R closed under x, $e_i \in R$

$$\lambda \cdot x = \sum_{i=1}^{5} e_i c_i \quad , e_i \in R$$

Thus it is expressible in linear combination

$\therefore \lambda x \in U$ (closed under scaler multiplication)

Hence $U \neq 0$, U is closed under (+) and ($\lambda \cdot$) remaining axioms are inherited from R^3. Hence col (C) is subspace of R^3

Base Selection Rule: Pivot of col of $\boxed{\text{original}}$ Matrix

$\therefore$ Basic (col (C)) $= \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} \right\}$

$\therefore$ dim (col (C)) = 3 $\Rightarrow$ Col (C) = R^3

Part ii) Row space Row

To prove $W = Row(C)$ is a subspace of R^5 and finding its dim & basis.

Step 1 : Subset property proof : $W \subseteq R^5$

Given $W = span(r_1, r_2, r_3) = \{ x \in R^3 : x = \sum_{i=1}^{3} a_i r_i \}$

$a_i \in R$, $V = R^5$

proof : Since we have each row vector ~~~~ $r_i \in R^5$
R^5 is closed under $(+)$, $(\cdot)$, any linear combination $x = \sum_{i=1}^{3} a_i r_i$ remains inside R^5

Thus $W \subseteq R^5$

Step 2 : To prove : $0 \in W \Rightarrow W \neq 0$

~~Assumption~~
proof :
we have any coefficient a_i & $a_1 = a_2 = a_3 = 0$

$\Rightarrow 0 \cdot r_1 + 0 r_2 + 0 r_3 = (0, 0, 0, 0, 0)$

$0 \in W \Rightarrow W \neq \phi$

proved

Step 3 & 4: Closure

① Addition since we have $x = \sum a_i r_i$ and
$y = \sum b_i r_i$.

$\Rightarrow \sum a_i r_i + \sum b_i r_i$
$\Rightarrow \sum (a_i + b_i) r_i \Rightarrow \sum d_i r_i$

where ~~$d_i \in R$~~ $d_i = a_i + b_i \in R$.

Thus proved $x + y \in W$

② Scalar Multi $\Rightarrow$ Since we have $x = \sum a_i r_i$, $\lambda x = \sum \lambda a_i r_i$

$\Rightarrow \lambda x = \sum_{i=1}^{5} e_i r_i$, where $e_i = \lambda a_i \in R$

Thus proved.

Hence we have $W \neq \phi$, W closed under $(+)$ & $(\lambda \cdot x)$
remaining axioms are inherit from R^5

Row (C) is subspace of R^5

Basis : Non zeros Row of $\boxed{REF}$ Matrix

Basis (Row(C)) = {(1, 2, 1, 2, -1), (0, -1, -1, -3, 8), (0, 0, 7, 11, -19)}

Dim (Row (C)) = 3

Sol^n 4 $\quad v_1 = (1,0,2)^T \quad v_2 = (1,2,2)^T \quad S_1 = \text{Span}\{v_1 \cdot v_2\}$

$w_1 = (1,1,0)^T, \quad w_2 = (0,1,1)^T \quad S_2 = \text{Span}\{w_1 \cdot w_2\}$

$S_1 = \text{span}\{v_1 \cdot v_2\} \subset R^3$
$S_2 = \text{span}\{w_1 \cdot w_2\} \subset R^3$

Part - 1 $\quad$ proove $S_1 \cap S_2$ is subspace of R^3

Step1: To proove $S_1 \cap S_2 \subseteq S_1 \subseteq R^3$

Given: $S_1; S_2$ are subspaces of R^3 (in problem given)

proof: since we have $x \in S_1 \cap S_2 \Rightarrow x \in S_1$
since $S_1 \subseteq R^3$, $x \in R^3$

Thus $S_1 \cap S_2 \subseteq R^3$.

Step2: Nonemptiness

To proove: $0 \in S_1 \cap S_2 \Rightarrow S_1 \cap S_2 \neq 0$

proof: since we have S_1 is a subspace $\Rightarrow 0 \in S_1$ &
S_2 is a subspace $\Rightarrow 0 \in S_2$

Thus $0 \in S_1 \cap S_2 \Rightarrow S_1 \cap S_2 \neq \phi$

Step 3 Closure under Vector Addition.

To prove : $x, y \in S_1 \cap S_2 \Rightarrow x+y \in S_1 \cap S_2$

Assumptions:

Let $x, y \in S_1 \cap S_2 \Rightarrow x, y \in S_1$ and $x, y \in S_2$

proof :

since we have x, y in both subspaces:

- In S_1: $x = a_1 v_1 + a_2 v_2$
  $y = b_1 v_1 + b_2 v_2$

$$x+y = (a_1 + b_1) v_1 + (a_2 + b_2) v_2$$
$$= d_1 v_1 + d_2 v_2 \in S_1$$

where $d_i = a_i + b_i \in R$

- In S_2: $x = a_3 w_1 + a_4 w_2$
  $y = b_3 w_1 + b_4 w_2$

$$x+y = (a_3 + a_3) w_1 + (b_3 + b_4) w_2$$

$$= d_3 w_1 + d_4 w_2$$

where $d_i = a_i + b_i \in R$

Thus proved :

$x+y \in S_1$ and $x+y \in S_2 \Rightarrow x+y \in S_1 \cap S_2$

Step 4: To prove $x \in S_1 \cap S_2$, $\lambda \in R \Rightarrow \lambda x \in S_1 \cap S_2$

Assumption: $x \in S_1 \cap S_2$ arbitrary & $\lambda \in R$

Proof:
Since we have x in both subspaces.

- In S_1 : $\lambda x = (\lambda a_1)v_1 + (\lambda a_2)v_2 = c_1 v_1 + c_2 v_2 \in S_1$
where $c_i = (\lambda a_i) \in R$

- In S_2 : $\lambda x = (\lambda a_3)w_1 + (\lambda a_4)w_1 = c_3 w_1 + c_4 w_2 \in S_2$

where $c_i = \lambda a_i \in R$

$\therefore$ Thus $\lambda x \in S_1$ and $\lambda x \in S_2 \Rightarrow \lambda x \in S_1 \cap S_2$

Hence: $S_1 \cap S_2 \neq 0$, closed under $(+)$ & $(\lambda \cdot)$

$\therefore S_1 \cap S_2$ is a subspace of R^3

Part (B) Finding Basis & dimn of $S_1 \cap S_2$

Step 5: Equate Parametrisation.
Set up homogenous set up $MZ = 0$
where $Z = \cancel{(a, b, c, d)}^T$ unknown vector

~~temporaries~~ $Mz = 0$ ~~for~~.

Since we have $x \in S_1 \cap S_2$ we will express x using spanning recipie

$$x = a_1 v_1 + \cancel{b} v_2 \cancel{+ c w_1 +}$$

$$x = c w_1 + d w_2$$

Equate both.

$$a_1 v_1 + b v_2 = c w_1 + d w_2$$

$$a_1 v_1 + b v_2 - c w_1 - d w_2 = 0$$

$$a \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + b \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - c \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - d \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Thus $Mz = 0$ found with $z = (a, b, c, d)^T$ unknown vector.

Step 6: Solve $Mz = 0$ via REF
find non trivial coeffic solⁿ.

$$M = [v_1 \; v_2 \; -w_1 - w_2]$$

$$M = \begin{bmatrix} 1 & 1 & -1 & 6 \\ 0 & 2 & -1 & -1 \\ 2 & 2 & 0 & -1 \end{bmatrix}$$

$R_3 \leftarrow R_3 - 2R_1$:

$$= \begin{pmatrix} 1 & 1 & -1 & 6 \\ 0 & 2 & -1 & -1 \\ 0 & 0 & 2 & -1 \end{pmatrix}$$

Basis variable : a, b, c
pivot = Rank = 3
Free variable = d = 1

Rank–Nullity Theorem:

$$\text{Nullity}(M) = \text{no. of } c - \text{rank}(M)$$
$$= 4 - 3 = 1 \text{ free variable } d.$$

Step 7 – Back substitution & Dual verification.

$R_3 = 2c - d = 0 \Rightarrow c = d/2$
$R_2 = 2b - c - d = 0 \Rightarrow b = 3d/4$
$R_1 = a + b - c = 0 \Rightarrow a = -d/4$

~~Lets d=4 let~~ let take $d = 4$, $a = -1$, $b = 3$, $c = 2$

Dual vector verification

- via S_1 : $x = -1(1,0,2)^T + 3(1,2,2)^T$
$= (2,6,4)^T$ ✓

- via S_2 : $x = 2(1,1,0)^T + 4(0,1,1)^T = (2,6,4)^T$
$= (\cancel{2+0\cdot 2+4, 0+4})^T$ ✓

Scaling down by 2 gives vector $u = (1, 3, 2)^T$

Thus vector $u = (1, 3, 2)^T \in S_1 \cap S_2$

Step 8 Basis and dimension calc^n.

Basis of $S_1 \cap S_2$:

① $\text{Basis}(S_1 \cap S_2) = \left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right\}$

② $\dim (S_1 \cap S_2) = \text{Nullity}(M) = 1$

Sol^n 5 $\quad v_1 = (1,1,0,0)^T \quad w_1 = (0,0,1,1)^T$

$v_2 = (0,1,1,0)^T \quad w_2 = (0,0,0,1)^T$

$S_1 = \text{Span}\{v_1, v_2\} \subset R^4$

$S_2 = \text{span}\{w_1, w_2\} \subset R^4$

Part(a) To prove S_1+S_2 is subspace of R^4

Step 1: $S_1+S_2 \subseteq R^4$ to prove.

Given:

$S_1+S_2 = \{p \in R^4 : p = x+y,\ x \in S_1,\ y \in S_2\}$

$V = R^4$

Since we have $x \in S_1 \subseteq R^4$ and $y \in S_2 \subseteq R^4$

R^4 is closed under $(+)$; $x+y \in R^4$

This prove $S_1+S_2 \subseteq R^4$

Step 2: To prove $0 \in S_1+S_2 \Rightarrow S_1+S_2 \neq \phi$

Since, S_1 ; S_2 are subspace $\Rightarrow 0 \in S_1$ and $0 \in S_2$

$0 = 0+0 \quad 0 \in S_1,\ 0 \in S_2$

Thus $0 \in S_1+S_2 \Rightarrow S_1+S_2 \neq 0$

Step 3 To prove

$p, q \in S_1 + S_2 \Rightarrow p+q \in S_1 + S_2$

Assumption: ~~...~~ let $p, q \in S_1 + S_2$, arbitrary

$p = x + y \; (x \in S_1 ; y \in S_2)$

$q = u + v \; (u \in S_1 , v \in S_2)$

Proof:

Since we have $p = x+y$ and $q = u+v$:

$p+q = (x+y) + (u+v) = (x+u) + (y+v)$

Since S_1 is a subspace; $x+u \in S_1$
Since S_2 is a subspace, ~~...~~ $y+v \in S_2$

Thus proved $p+q \in S_1 + S_2$ (closed(+))

Step 4
To prove: $p \in S_1 + S_2 ; \lambda \in R \Rightarrow \lambda p \in S_1 + S_2$

Assumption:

Let $p = x + y$ ($x \in S_1$, $y \in S_2$) and $\lambda \in R$

proof:

Since we have $p = x + y$.

$$\lambda p = \lambda x + \lambda y.$$

Since S_1 is subspace; $\lambda x \in S_1$.

Since S_2 is subspace, $\lambda y \in S_2$.

Thus proved.

$\lambda p \in S_1 + S_2$ (Closed under scalar multiplication)

Hence

i) $S_1 + S_2 \neq 0$

ii) Closed under Addition

iii) Closed under Scalar Multiplication

iv) Remaining Axioms are inherit from R^4

$\therefore$ $S_1 + S_2$ is subspace.

Part b) Basis & dim of $S_1 + S_2$

Step 5 : Pool spanning vector into Matrix M

~~To prove~~ : Form Matrix M all spanning vector of S_2 & S_2 as colm.

$$M = (v_1\ v_2\ w_1\ w_2) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Step 6 : Reduce to REF count pivot

1. Apply $R_2 \leftarrow R_2 - R_1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

2. Apply $R_3 \leftarrow R_3 - R_1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

3. Apply $R_4 \leftarrow R_4 - R_3$ $\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$Rank(M) = 4$

Step 7. Basic & Dim cal^n.

Basic $(S_1 + S_2)$ = Pivot column of original Matrix

$$= \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Dimension : $\dim(S_1 + S_2) = 4$

Step 8 : Dimension Formula verificat^n.

$$\dim(S_1 + S_2) = \dim S_1 + \dim S_2 - \dim(S_1 \cap S_2)$$

$$= 2 + 2 - 0 = 4$$

L.H.S = R.H.S.